Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.

IS : 4758 - 1968

(Reaffirmed 2002)

(Reaffirmed 2012)

(Reaffirmed 2017)

Indian Standard

METHODS OF MEASUREMENT OF NOISE EMITTED BY MACHINES

(Fourth Reprint DECEMBER 1995)

UDC 534.837

© *Copyright* 1968

BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

Gr 4 *December* 1968

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.93.161

Indian Standard

METHODS OF MEASUREMENT OF NOISE EMITTED BY MACHINES

Acoustics Sectional Committee, ETDC 27

Chairman	*Representing*
DR M. PANCHOLY	National Physical Laboratory (CSIR), New Delhi
Members	
SHRI A. N. AHUJA	Ahuja Radios, New Delhi
SHRI I. S. AHUJA (*Alternate*)	
SHRI G. S. BOLINA	G. W. Balkar & Co, Jullundur
SHRI S. C. ISRANI	The Radio Electronics & Television Manufacturer's Association, Calcutta
SHRI R. K. JAIN (*Alternate*)	
SHRI H. JOGA RAO	State Broadcasting Department, Government of Andhra Pradesh
SHRI B. G. KALE	Railway Board (Ministry of Railways)
LT-COL S. D. MAINI	Directorate General of Inspection (Ministry of Defence)
SHRI P. KRISHNAMURTHY RAO (*Alternate*)	
SHRI G. B. MEEMAMSI	Posts & Telegraphs Department, New Delhi
SHRI J. S. MONGA	All India Radio Merchants' Association, Bombay
SHRI V. J. BHATT (*Alternate*) (Bombay)	
SHRI D. N. CHAUDHURI (*Alternate*) (Calcutta)	
SHRI L. S. V. EASWAR (*Alternate*) (Madras)	
SHRI S. N. MUKERJI	National Test House, Calcutta
SHRI B. P. GHOSH (*Alternate*)	
SHRI M. S. NARAYANAN	Directorate General of Defence Research & Development (Ministry of Defence)
DR B. S. RAMAKRISHNA	Indian Institute of Science, Bangalore
RESEARCH ENGINEER	Directorate General of All India Radio
DR (SMT) C. K. KESAVAN (*Alternate*)	
SHRI M. SANKARALINGAM	Directorate General of Supplies & Disposals (Inspection Wing)
DR D. L. SUBRAHMANYAM	Central Electronics Engineering Research Institute (CSIR), Pilani
SHRI K. D. PAVATE (*Alternate*)	
SHRI Y. S. VENKATESWARAN, Director (Elec tech)	Director General, ISI (*Ex-officio Member*)

Secretaries

SHRI N. SRINIVASAN
Deputy Director (Elec tech), ISI

SHRI S. MUKHOPADHYAY
Assistant Director (Elec tech), ISI

BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - RAT(naikassociates110@gmail.com) 116.74.95.151.

Indian Standard

METHODS OF MEASUREMENT OF NOISE EMITTED BY MACHINES

0. FOREWORD

0.1 This Indian Standard was adopted by the Indian Standards Institution on 3 September 1968, after the draft finalized by the Acoustics Sectional Committee had been approved by the Electrotechnical Division Council.

0.2 This standard concerns with the procedures followed in the objective measurements of noise emitted by machines. These procedures are not necessarily applicable to noise of an impulsive character.

0.3 This standard describes the general principles of methods of measurements of machinery noise based on which individual standards may be formulated. The specific standard covering methods of measurement of noise of the various types of machines will have to select the most suitable method having regard to the size of the machine and its application. All the necessary particulars to enable a result to be obtained with the required accuracy will be given in these standards.

0.4 The measurement of the physical characteristics of the noise of machines has the following four objects:

a) To verify that the noise of a given machine conforms to a certain standard,

b) To make a comparison between the noise emitted by machines built to the same specification,

c) To make a comparison between the noise emitted by different machines, and

d) To determine the noise received at a distance.

The four aims specified above are achieved by determining the acoustic noise power radiated by the machine as well as the directivity, these quantities being measured as a function of the frequency. Determination of the acoustic power should thus be the preferred method for the measurement of the noise of machines. The different methods of measuring acoustic power are given in Appendices A to C.

0.4.1 Determination of the power and the directivity, however, leads to the siting of the points of measurement sufficiently removed from the machine for the conditions of propagation by progressive waves to be established. It is also necessary that at this site, the reflections from the walls of the premises shall have no effect. This requires premises of relatively large dimensions compared with those of the machine, with very absorbent walls.

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

0.4.2 It is sometimes possible to use rooms with very reverberant walls, where the acoustics field tends to become uniform. But the directivity of the noise radiated cannot be determined under these conditions. In addition, the problems of the special premises remain the same.

0.4.3 It may also be possible to use semi-reverberant spaces but this method can provide only the limited information on the directivity of acoustic radiation and very rough estimation of the acoustic power.

0.4.4 While there will be no difficulty to provide necessary acoustic environment for small machines, it may not always be possible to achieve the necessary conditions for large machines, special installation condition of the machine may be required. These conditions cannot always be met particularly for large machines. It may not also be possible to bring the machine to a room suitable for the determination of the acoustic power. Hence, use is often made of a simplified method in which sound pressure measurements are carried out at certain measurement points suitably spread around the outer casing of the machine, sufficiently close to the machine to obviate the effect of the sound reflections. Such a method enables the object (a) to be achieved with the desired precision, the dimensions of the machine being generally sufficiently well-defined, but it does not enable objects (c) and (d) to be obtained. It is, however, possible for the acoustic conditions at a distance to be evaluated approximately on the basis of pressure measurements carried out in the vicinity of the machine by taking into account the surface on which the measurements have been carried out.

0.5 This standard follows closely ISO Recommendation 495 'General requirements for the preparation of test codes for measuring the noise emitted by machines' issued by the International Organization for Standardization.

0.6 In reporting the result of a test made in accordance with this standard, if the final value, observed or calculated, is to be rounded off, it shall be done in accordance with IS : 2-1960*.

---

1. SCOPE

1.1 This standard deals with the procedures to be followed in the objective measurement of the noise emitted by machines. These procedures are not necessarily applicable to noise of an impulsive character.

2. TERMINOLOGY

2.0 For the purpose of this standard, the following definitions shall apply.

*Rules for rounding off numerical values (*revised*).

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

2.1 Sound Pressure Level — Twenty times the common (Briggsian) logarithm of the ratio of a given sound pressure to be stated reference pressure.

NOTE 1 — Unless otherwise explicitly stated, it is to be understood that the sound pressure is the effective (rms) sound pressure.

NOTE 2 — It is to be noted that in many sound fields the sound pressure ratios are not the square roots of the corresponding power ratios.

2.2 Sound Level — The reading given by a sound level meter complying with the relevant Indian Standard (*see* IS : 3932-1966*).

2.3 Sound Power Level — Ten times the common (Briggsian) logarithm the ratio of a given sound power to a stated reference power.

2.4 Machine — Any sound source of which the acoustical characteristics are to be measured.

2.5 Prescribed Surface — A hypothetical surface surrounding the machine as given in the individual standard and over which the measurements are assumed to be made. Its area is calculated as laid down in the individual standard.

2.6 Equivalent Hemisphere — Hypothetical hemisphere having the same area as the prescribed surface.

2.7 Reference Radius — Radius (measured from the centre of the equivalent hemisphere), as given in the individual standard, to which all the results of measurements made on machines of the same category (tested according to the same standard), are reduced.

3. GENERAL

3.1 Installation and Operation of the Machine — The acoustic radiation of the machine can depend on its installation, particularly when the machine is one of small dimensions. The condition of installation and the conditions of operation of the machine during the test shall be included in the individual specification.

3.2 Quantities to be Measured — Depending on the purpose of the measurement, nature of the noise source, and the character of the sound field, one or both of the following quantities should be measured:

a) Sound level (it is recommended that the weighting network A will be adopted), and

b) Sound pressure level in frequency bands of given width.

NOTE — It is recommended that octave bands be used. For narrower bands, the result should be expressed in octave band.

---

*Specification for sound level meters for general purposes use.

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

3.2.1 Since in all cases a pressure microphone will be used, the result of the measurements will be band sound pressure level or sound level values.

NOTE 1 — When the spectral distribution of the noise is well known, a measurement of wide band sound pressure level or sound level may suffice.

NOTE 2 — Any determination of the spectral distribution of the noise should be in frequency bands corresponding to those specified in IS : 2264-1963*.

3.3 Measuring Equipment — A sound level meter (*see* IS : 3932-1966†) or a frequency analyzer or both shall be used as specified in the relevant standard.

3.3.1 The measuring equipment shall be calibrated against acoustic standards at suitable intervals.

3.4 Measuring Positions — In the individual standard, the sound pressure level measuring positions are to be clearly laid down.

4. METHODS OF MEASUREMENT

4.1 The accurate determination of the acoustic power of a machine (*see* **0.4**) has to be performed either as a far-field sound pressure level measurement under hemispherical radiation or free-field conditions, or as a sound pressure level measurement in a reverberant room under diffuse field conditions. In cases where it is difficult to take sound pressure level measurements under these conditions, the measurements may have to be performed under semi-reverberant conditions (*see* Appendix C), or near-field measurements (*see* Appendix D) be made.

4.1.1 The following four methods are available for the determination of acoustic noise power of the machine and/or the sound pressure level of the noise emitted:

a) *Hemispherical radiation and free-field methods* — The basic principle of this method and evaluation of the radiated acoustic noise power and its directivity are described in Appendix A;

b) *Diffuse-field method* — The basic principle of this method and evaluation of acoustic noise power are described in Appendix B. This method does not provide information on the directivity of the acoustic radiation. Since the degree of diffusion is dependent upon the nature of the noise source, this method is not recommended for machines the noise from which contains prominent components;

c) *Semi-reverberant method* — The basic principle of this method is described in Appendix C. This method provides only limited information on the directivity of the acoustic radiation and a rough calculation of radiated acoustic noise power; and

*Preferred frequencies for acoustical measurements.

†Specification for sound level meters for general purposes use.

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

d) *Near-field method* — In this method the sound pressure level measurements only are made close to the machines. Details of this method along with the required calculations and evaluation of approximate acoustic noise power are given in Appendix D. This method is particularly applicable to large machines (*see* **0.4**).

4.2 Correction for Background Noise — The background noise readings when the machine is not on test should be determined using the same filters and at the same points as for the test. The readings at each point with the machine on test ought to exceed those due to the background noise alone by at least 10 dB.

4.2.1 When the differences are less than 10 dB, corrections as given below shall be applied:

dB Increase in Level Produced by the Machine (dB)	dB to be Subtracted from the Measured Value (dB)
3	3
4-5	2
6-9	1

When corrections are applied the corrected levels should be reported in brackets.

4.2.2 When the increase is less than 3 dB, measurements in general cease to have any significance.

5. PRESENTATION OF TEST RESULTS

5.1 The test report shall give the following information in all cases:

a) Reference to individual standard, if any;

b) Description of the machine and of its conditions of installation and operation;

c) Description of the test environment and location of the machine;

d) Meteorological conditions, if appropriate, for example, ambient temperature, relative humidity and barometric pressure;

e) Description of measuring apparatus used;

Note 1 — When a sound level meter is used, the type employed and the weighting network used should be stated.

Note 2 — When a frequency analyzer is used, the band widths and centre frequencies should be stated.

f) Position of measuring points;

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151

g) Results of sound pressure level measurements;

h) Background noise levels;

j) Sound pressure level values corrected for background noise, if necessary:

k) When required, the calculated octave band sound power levels;

m) When required, the sound pressure level at the reference radius or reference surface corrected as for free-field conditions; and

n) In the case of measurements in an anechoic environment, when required, the directivity index as calculated by the method of Appendix A.

NOTE — Values of the directivity index at intervals of 30° are generally sufficient.

5.1.1 *Additional Information to be Given for Near-Field Sound Pressure Level Measurements*

a) Description of the prescribed surface and equivalent hemisphere.

b) Results of sound pressure level measurements, either average or individual, as specified in the individual standard.

c) Effect, if any, of environment.

d) Extrapolation of the results to the reference radius specified in the individual standard by the method given in Appendix D.

6. CHOICE OF METHOD

6.1 The objects specified in **0.4** are achieved by determining the acoustic power radiated by the machine as well as the directivity. Hence the method that permits the determination of acoustic power is recommended. However, the choice of method depends on the size of the machine and its application.

6.1.1 The methods specified in **4.1.1** (a) enable the determination of the acoustic noise power of the machine and its directivity (physical properties of the machine noise) and the calculation of the sound pressure level in other distance or other rooms.

If the directivity is not pronounced or there are no great variations from one machine to another, it may be preferable to determine the acoustic power from measurements in a diffuse field as specified in **4.1.1** (b). In this case since the characteristics of the reverberant enclosure generally vary with frequency, it is usually necessary to analyze the noise in frequency bands.

These two methods are preferred methods.

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - naikassociates110@gmail.com) 116.74.95.151.

6.1.2 In certain cases, the method specified in **4.1.1** (c) which is less accurate may be acceptable. Determination of acoustic power by this method is often less precise.

6.1.3 When measurements according to **4.1.1** (a) to (c) are difficult to perform, measurement of the sound pressure level close to the machine and where appropriate, the determination of sound pressure level at the reference radius, in accordance, with **4.1.1** (d) may have to be made. These may be the only measurements possible when difficulties are encountered with reflecting surfaces or background noise. Such measurements permit correct comparisons to be made between similar machines built to the same specifications but make possible only an approximate determination of the acoustic power or the sound pressure level at a distance.

APPENDIX A

[*Clause* 4.1.1 (a)]

DETERMINATION OF ACOUSTIC POWER AND DIRECTIVITY INDEX UNDER HEMISPHERICAL RADIATION AND FREE-FIELD CONDITIONS

A-1. PRINCIPLE OF MEASUREMENT

A-1.1 The determination of the power and the directivity requires the siting of measurements points sufficiently remote from the machine for the conditions of propagation of progressive sound waves to be established. There should be no reflected waves at the measurement points and this requires premises with sufficiently absorbent walls and dimensions large compared with those of machine. Any effect of the surroundings or near-field effects will be made apparent by measuring over two surfaces and checking if the sound pressure varies inversely as the square root of the ratio of surface areas. A ratio of areas of 2 or 4 is recommended, corresponding to −3 and −6 dB respectively. The results of such check measurements should be stated.

A-2. DETERMINATION OF ACOUSTIC POWER

A-2.1 Many machines are mounted on or near a reflecting floor plane. Under these conditions, in the absence of reflections from other surfaces of the environment and in the absence of absorption in the surroundings, the sound power level may be determined. A hemisphere is marked out around the machine and the sound power level calculated as follows:

$$10 \log_{10} \frac{P}{P_0} = 20 \log_{10} \frac{p_m}{p_0} + 10 \log_{10} \frac{2\pi r^2}{S_0}$$

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

In this equation:

$10 \log_{10} \frac{P}{P_0}$ is the sound power level of the machine under test, above,

$$P_0 = 10^{-12}\ W$$

$20 \log_{10} \frac{p_m}{p_0}$ is the mean sound pressure level, above

$$p_0 = 2 \times 10^{-5}\ \text{N/m}^2\ (2 \times 10^{-4}\ \text{dyn/cm}^2)$$

$2\pi r^2$ is the surface of the test hemisphere, relative to $S_0 = 1\ \text{m}^2$.

The quantity p^2_m is the mean value in space of the square of the rms sound pressures recorded at a certain number of measuring points, on a test hemisphere of radius r with the machine as its centre. It is necessary that r be sufficiently large for the measuring points to come within the field of progressive waves where sound pressure and particle velocity are in phase. In the most usual cases, a value of r of the order of twice the dimensions of the machine suffices to fulfil this condition. In the above expression the acoustic impedance of the air is assumed to be equal to $\rho C = 400$ mks (rayls). When the machine is not mounted on a reflecting plane but is in free space, the radiation becomes spherical and not hemispherical and $2\pi r^2$ should be replaced by $4\pi r^2$.

A-3. DETERMINATION OF THE DIRECTIVITY INDEX

A-3.1 The directivity index can be calculated as follows on the basis of the measurements carried out under hemispherical radiation conditions:

$$DI = 20 \log_{10} \frac{p}{p_0} - 20 \log_{10} \frac{p_m}{p_0} + 3\ \text{dB} \qquad \ldots(2)$$

In this equation :

DI is the directivity index in the direction considered, the reflecting plane being regarded as an integral part of the source.

$20 \log_{10} \frac{p}{p_0}$ is the sound pressure level, above

$$p_0 = 2 \times 10^{-5}\ \text{N/m}^2\ (2 \times 10^{-4}\ \text{dyn/cm}^2)$$

at the distance from the centre equal to r in the direction considered

$20 \log_{10} \frac{p_m}{p_0}$ is the mean sound pressure level, above

$$p_0 = 2 \times 10^{-5}\ \text{N/m}^2\ (2 \times 10^{-4}\ \text{dyn/cm}^2)$$

on the test hemisphere of radius r.

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

When the machine is not mounted on a reflecting plane but is in free space, the radiation becomes spherical and not hemispherical and the 3 dB in equation 2 should be omitted.

APPENDIX B

[*Clause* 4.1.1. (b)]

DETERMINATION OF ACOUSTIC POWER UNDER DIFFUSE FIELD CONDITIONS

B-1. PRINCIPLE OF MEASUREMENT

B-1.1 As an alternative to the hemispherical radiation and free-field methods, the use of a highly reverberant enclosure is recommended. In this case, the sound field tends to become uniform and the relevant theory for sound power determination is described in **B-2**. This method can provide no information on the directivity of the acoustic radiation.

B-2. DETERMINATION OF ACOUSTIC POWER

B-2.1 If the machine under test is placed on the floor of a reverberant room of suitable shape, and having dimensions and reverberation time such that the sound field is appreciably diffuse throughout most of the room, the sound power P can be calculated as follows:

$$10 \log_{10} \frac{P}{P_0} = 20 \log_{10} \frac{p_m}{p_0} - 10 \log_{10} \frac{T}{T_0} + 10 \log_{10} \frac{V}{V_0} - 14 \text{ dB} \qquad \ldots(3)$$

In this equation:

$10 \log_{10} \frac{P}{P_0}$ is the sound power level of the machine under test, above

$$P_0 = 10^{-12} \text{ W}$$

$20 \log_{10} \frac{p_m}{p_0}$ is the mean sound pressure level, above

$p_0 = 2 \times 10^{-5}$ N/m² (2×10^{-4} dyn/cm²)

T = the reverberation time in seconds

T_0 = 1 second

V = the volume of the room in m³

V_0 = 1 m³

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

The quantity p^2_m is the mean square sound pressure obtained from measurements at a certain number of points in the diffuse (reverberant) field in the room. If it is not possible or convenient to determine the reverberation time T, P can be determined by using a reference sound source of known acoustic power (*see* Appendix C).

NOTE 1 — The microphone shall not be placed too near the boundries of the room. A minimum distance of $\frac{\lambda}{4}$ shall be maintained, where λ is the wavelength of the sound at the lowest frequency of interest.

NOTE 2 — The microphone shall be placed sufficiently away from the source to be in the reverberant field and not in the near field of the source. A minimum distance of $\frac{2}{3} V^{\frac{1}{3}}$ is recommended, where V is the volume of the room.

NOTE 3 — The microphone positions shall be selected to cover approximately equal surface area over the test hemisphere (sphere).

APPENDIX C

[*Clause* 4.1.1(c)]

DETERMINATION OF SOUND POWER UNDER SEMI-REVERBERANT CONDITIONS

C-1. PRINCIPLE OF MEASUREMENT

C-1.1 In practice, it is often necessary for machines to be tested in room having limited absorption (semi-reverberant rooms), such as factories, shops, offices. Under these conditions, the sound power of the machine under test can be determined by using a reference sound source of known acoustic power P_r which is substituted for the machine. If it is not possible for the machine under test to be moved, and if the test room is large enough, another suitable position for the reference source can usually be found. The reference sound source can be a machine of similar type or a special sound source radiating noise having a sufficiently uniform spectral distribution over the whole frequency range of interest. Determination of the sound power of the reference source is to be carried out by the method described in Appendix A and Appendix B, or the result can also be provided by a testing laboratory.

C-2. DETERMINATION OF ACOUSTIC POWER

C-2.1 The acoustic power level P of the machine under test can be calculated as follows:

$$10 \log_{10} \frac{P}{P_o} = 10 \log_{10} \frac{P_r}{P_o} + 20 \log_{10} \frac{p_m}{p_o} - 20 \log_{10} \frac{p_{mr}}{p_o} \qquad \ldots(4)$$

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

In this equation:

$10 \log_{10} \frac{P}{P_0}$ is the sound power level of the machine under test, above

$$P_0 = 10^{-12} \text{ W}$$

$10 \log_{10} \frac{P_r}{P_0}$ is the sound power level of the reference source, above

$$P_0 = 10^{-12} \text{ W}$$

$20 \log_{10} \frac{p_m}{p_0}$ is the mean sound pressure level of the machine under test, above

$$p_0 = 2 \times 10^{-5} \text{ N/m}^2 \ (2 \times 10^{-4} \text{ dyn/cm}^2)$$

$20 \log_{10} \frac{p_{mr}}{p_0}$ is the mean sound pressure level of the reference source, above

$$p_0 = 2 \times 10^{-5} \text{ N/m}^2 \ (2 \times 10^{-4} \text{ dyn/cm}^2)$$

The quantities p^2_m and p^2_{mr} are the space averages of the mean square sound pressures recorded at a certain number of points on the test hemisphere of radius r in the semi-reverberant room, with the machine as its centre. The radius r is chosen as in the case for hemispherical radiation or free-field conditions (Appendix A). The measuring points are chosen in the same way for the reference source and for the machine under test.

C-2.2 In many cases a region of the sound field can be found where the sound pressure level beyond the distance r does not vary appreciably with position. In that event, the above equation can still be used to calculate P. The quantities p^2_m and p^2_{mr} then represent the mean values of the squares of the rms sound pressures recorded at a certain number of points in the reverberant field of the test room. This procedure can also be followed in a reverberant room (*see* Appendix B).

APPENDIX D

[*Clause* 4.1.1 (d)]

NEAR-FIELD SOUND PRESSURE LEVEL MEASUREMENT

D-1. MEASUREMENT

D-1.1 Prescribed Surface — A surface as simple as possible and whose area can be calculated easily is to be marked out round the machine conforming approximately to the external casing. This surface and its

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

average distance from the outer casing are to be laid down in each individual standard. The number and disposition of the measuring points required depend on the irregurlarity of the acoustic field. When it is necessary to carry out measurements closer to the machine than is specified in the individual standard, or to explore only part of the surface, these limitations should be clearly stated in the test report.

D-1.2 Evaluation of the Result — In the majority of cases, in particular when the directivity is not pronounced or when for the type of machine under consideration it varies relatively slightly from one machine to another, the arithmetic mean of the decibel readings over the prescribed surface may be a sufficiently accurate measure of the mean square pressure.

NOTE — For a spread of 5 dB in the decibel readings the error in taking the mean of the decibels is less than 0·7 dB and for a spread of 10 dB it is less than 2·5 dB. When the spread is greater than 10 dB, this method of averaging is generally not suitable.

D-1.3 Effect of Environment — The effect of the surroundings of the machine may be determined by first placing such a machine in a space without reflections and secondly in the test environment so that a direct comparison can be made; every precaution should be taken to ensure constant conditions of machine noise and measuring apparatus. As accurate determination of the influence of the surroundings is not possible it is recommended that this influence be kept to a minimum.

D-2. CALCULATION OF THE SOUND PRESSURE LEVEL AT THE REFERENCE RADIUS

D-2.1 The sound pressure level at a reference radius which is laid down in the specific individual standards is calculated from that measured on the prescribed surface as described in **D-2.2.** The reference radius is to be laid down in each individual standard. It should be the same of all machines covered by a given standard. One of the values 1, 3 or 10 m should be adopted. It is recommended that the value chosen should be slightly greater than the radius of the equivalent hemisphere corresponding to the largest machine covered by the individual standard.

D-2.2 The free-field sound pressure level at the reference radius is to be calculated thus:

$$20 \log_{10} \frac{p_d}{p_o} = 20 \log_{10} \frac{p_m}{p_o} - 20 \log_{10} \frac{d}{r}$$

$$p_o = 2 \times 10^{-5} \text{ N/m}^2 \ (2 \times 10^{-4} \text{ dyn/cm}^2)$$

where

$$r = \sqrt{\frac{S}{2\pi}} = \text{radius of equivalent hemisphere,}$$

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

S = area of prescribed surface,

d = reference radius,

p_m = mean sound pressure over the prescribed surface, and

p_d = sound pressure at the reference radius.

D-3. APPROXIMATE EVALUATION OF THE SOUND POWER

D-3.1 In certain cases it is possible to arrive at an approximate evaluation of the sound power starting from the sound pressure level at the measuring distance or the reference radius (*see* **D-2** and Appendix A).

NOTE — In the case of hemispherical radiation, the numerical value of sound power level would be 8, 18 or 28 dB greater than the sound pressure level at the reference distances of 1, 3 and 10 m respectively.

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.05.151

BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, NEW DELHI 110002

Telephones: 331 01 31, 331 13 75 — Telegrams: Manaksanstha (Common to all Offices)

Regional Offices:	*Telephone*
Central : Manak Bhavan, 9 Bahadur Shah Zafar Marg, NEW DELHI 110002	331 01 31 331 13 75
*Eastern : 1/14 C. I. T. Scheme VII M, V. I. P. Road, Maniktola, CALCUTTA 700054	36 24 99
Northern : SCO 445-446, Sector 35-C, CHANDIGARH 160036	2 18 43 3 16 41
Southern : C. I. T. Campus, MADRAS 600113	41 24 42 41 25 19 41 29 16
†Western : Manakalaya, E9 MIDC, Marol, Andheri (East), BOMBAY 400093	6 32 92 95
Branch Offices:	
'Pushpak', Nurmohamed Shaikh Marg, Khanpur, AHMADABAD 380001	2 63 48 2 63 49
‡Peenya Industrial Area 1st Stage, Bangalore Tumkur Road BANGALORE 560058	38 49 55 38 49 56
Gangotri Complex, 5th Floor, Bhadbhada Road, T. T. Nagar, BHOPAL 462003	6 67 16
Plot No. 82/83, Lewis Road, BHUBANESHWAR 751002	5 36 27
53/5, Ward No. 29, R.G. Barua Road, 5th Byelane, GUWAHATI 781003	3 31 77
5-8-56C L. N. Gupta Marg (Nampally Station Road), HYDERABAD 500001	23 10 83
R14 Yudhister Marg, C Scheme, JAIPUR 302005	6 34 71 6 98 32
117/418 B Sarvodaya Nagar, KANPUR 208005	21 68 76 21 82 92
Patliputra Industrial Estate, PATNA 800013	6 23 05
T.C. No. 14/1421, University P.O., Palayam TRIVANDRUM 695035	6 21 04 6 21 17
Inspection Offices (With Sale Point):	
Pushpanjali, First Floor, 205-A West High Court Road, Shankar Nagar Square, NAGPUR 440010	2 51 71
Institution of Engineers (India) Building, 1332 Shivaji Nagar, PUNE 411005	5 24 35

*Sales Office in Calcutta is at 5 Chowringhee Approach, P. O. Princep Street, Calcutta 700072 — 27 68 00

†Sales Office in Bombay is at Novelty Chambers, Grant Road, Bombay 400007 — 89 65 28

‡Sales Office in Bangalore is at Unity Building, Narasimharaja Square, Bangalore 560002 — 22 36 71

Reprography Unit, BIS, New Delhi, India